Proceedings of the 9th International Conference on Electronics, Communication and Aerospace Technology (ICECA-2025)
IEEE Xplore Part Number: CFP25J88-ART; ISBN: 979-8-3315-9929-4

2025 9th International Conference on Electronics, Communication and Aerospace Technology (ICECA) | 979-8-3315-9929-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICECA66444.2025.11383295

Advanced Securing Cloud Environments using Quantum Convolutional Neural Network-based Multi-Channel Cryptographic

Kiran Kumar T. M[1], Chirag Mavani[2], Hirenkumar Mistry[3], Chintan thacker[4], Shirish, Ramya Maranan[5],Dr.M.SivaramKrishnan[6]
[1] Assistant Professor, Department of Computer Science and Engineering, Siddaganga Institute of Technology, Tumkur, Karnataka, 572102, India tmkiran@sit.ac.in
[2] Cloud/DevOps & Cybersecurity Engineer, 3165 Deodara St, Fremont, CA 94538, USA chiragmavanii@gmail.com
[3] Sr. Linux Administrator & Cloud Engineer, 1084 N Abbott Ave., Milpitas, CA 95035, USA hiren_mistry1978@yahoo.com
[4] Assistant Professor, Computer science and Engineering, Faculty of Engineering and Technology, Parul institute of Engineering and Technology, Parul University, Post Limda, Waghodia, Gujarat 391760, India chintan.thacker19435@paruluniversity.ac.in
[5] Department of Research and Innovation, Saveetha School of Engineering, SIMATS, Chennai, Tamil Nadu, 602105, India ramyamaranan@yahoo.com
[6]Associate Professor, Department of Electrical and Electronics Engineering, Karpagam College of Engnieering,India-krishbe95@gmail.com

Abstract*- The rapid proliferation of Internet of Things (IoT) devices has led to an unprecedented aggregation of sensitive information within cloud infrastructures, consequently accentuating vulnerabilities associated with confidentiality, data integrity, and infrastructural resilience. Furthermore, extant privacy frameworks not only face imminent threat from forthcoming quantum decryption capabilities but also exhibit inherent weaknesses attributable to centralized data architectures. In order to confront this intricate and evolving risk landscape, we present a multi-layered security architecture that integrates heuristic anomaly detection, countermeasures against inference-based and anomaly attacks, post-quantum encryption, and a distributed ledger architecture—thereby establishing a coherent and uniform defensive paradigm. Suspicious transaction detection at the initial screening stage relies upon the identification of multi-canonical anomalous transactions, applied through a Multi-Channel Quantum Convolution Neural Network. This core mechanism is synergistically enhanced by a Walk-Spread search algorithm, which prioritises the identification of anomalous transaction patterns across temporal scales mediated by quantum temporal frameworks, effectively strengthening detection capabilities. Recorded transaction payloads within the proposed system are secured through the McEliece post-quantum encryption scheme, thereby constructing a formidable cryptographic bulwark against a wide array of adversarial efforts, including those mounted by both classical and quantum computational means. Building upon this foundation, a streamlined adaptation of Practical Byzantine Fault Tolerance (PBFT) is engineered, realising a declarative and resource-efficient blockchain architecture that retains key attributes of consensus, immutability, cryptographic tamper-resistance, and thoroughly decentralised ledger operation while maintaining a and of very modest persistent computational cost. Evaluation against realistically emulative workload parameters demonstrates an anomaly detection accuracy exceeding 99%, with both precision and F1-score metrics achieving consensus at the operational maximum, collectively corroborating the system's resilience and effective adherence to operational metrics across contend time horizon.

Keywords— Cloud Computing, Security, Multi-Channel Quantum Convolutional Neural Network, McEliece Post-Quantum Cryptosystem, Walk-Spread Algorithm.

I. INTRODUCTION

Cloud computing constitutes the paramount infrastructural substrate for contemporary digital systems by delivering the levels of storage, computational, and distributive capacity necessitated by the constant and contextually undetectable streams of data generated by the Internet of Things (IoT). The rapid and, in many instances, automatic transition to serverless and microservices architectures seeks to minimize the latency between deployment and anomaly detection, yet this compression widens the cumulative storage and in-transit corpus of data exposed to breaches of confidentiality, integrity, and availability (CIA) [1]. The risks magnify within cloud infrastructures tethered to federated federated identity frameworks, where numerous, geographically disparate actors routinely transmit sensitive telemetry whose processing must be column-wide, temporally bound to compute units [2]. By rendering interactive planes as implicit and monopolized, they crystallize singular, and consequently attackable, lattice nodes as operational singular, and thus catastrophic, nodes of existential threat.

Enduring spikes in control-plane interface throughput disproportionately extend the operational range of denial-of-service, advanced-persistent-threat, and malicious adjunct-authentication vectors beyond established modelling horizons [3]. Such an empirical observation compels the immediate adoption of post-quantum cryptographic suites and self-reorganising trust-voting lattice structures, effecting the attenuation of compromise and sabotage vectors in the IoT–cloud federation milieu [4]. Concomitantly, the deliberate, stratified infusion of consensus-substrate paradigms into control architecture materially elevates the tamper-evidence guarantees of federated data-governance, quashing adversarial data counter-claims and instilling self-validating, provenance-agnostic trust levels [5],[6]. In systematic response, the present paper delineates an architecture–Multi-Channel Quantum Convolutional Neural Network invoking the Walk-Spread Algorithm (M-CQCNNet-WSA)–which amalgamates quantum-lattice, longevity-resilient keying primitives, an adaptive-traffic, path-diverse security-breach detector, and an anchored, trust-minimized long-term federated audit layer, thereby configuring a storable and operationally elastic safeguard architecture inherently engineered for expansive, federated IoT–cloud ecosystems [7].

Novelty and contribution

- **Quantum-Resilient Encryption:** Introduces Presents the McEliece Post-Quantum Cryptosystem as a means of securing the transmission of IoT data and strong protection against future quantum attacks.
- **Multi-Channel Quantum Deep Learning:** Designs a Multi-Channel Quantum Convolutional Neural Network (M-CQCNNet) which considers the laws of quantum mechanics on multiple channels of data that resulted in effective and efficient detection of anomaly.
- **Walk-Spread Algorithm Optimization:** Incorporates the Walk-Spread Algorithm (WSA) to tune the weights of neural networks in an optimal manner and increase the performance of detection with reductions of computational factors.
- **Blockchain-Based Secure Storage:** Introduces a Practical Byzantine Fault Tolerance Consensus-based Lightweight Blockchain (PBFT-CLBC) of decentralized, tamper-resistant and transparent store of IoT data that is led to the processed and stored in cloud environments.

The overall structure of the research study is organized as follows: Section 2 summarizes the most recent research on the topic, Section 3 discusses the proposed method in greater detail, and Section 4 shows results from the extended methodology and last Section 5 contains explanations of the study results.

II. LITERATURE SURVEY

The most recent research on security is summarized below,

In [8] presented a Scalable and Secure Cloud Architecture (SSCA) in 2023 that supports multi-user cloud settings by combining blockchain, cryptography, and IoT. The architecture uses the Multicast and Broadcast Rekeying Algorithm (MBRA) and Post-Quantum Cryptography (PQC) for improved security, and it has decentralized cloud nodes for quicker request processing. MBRA-PQC encrypts data generated by the Internet of Things, when blockchain provides data storage that can't be tampered with. Indicators for performance were security, scalability and response time. The results were better data protection, better key management, and better scalability. However, the performance of real time application might be affected by the added latency and processing burden of PQC and blockchain.

In [9] proposed integrating Explainable Deep Learning with a Blockchain-based consensus mechanism to transform cyber event detection and response. The increased means of real-time situational threat recognition by providing interpretability (via Explainable AI), and the ensuring of secure and shareable decisions (using blockchain), has improved the ability of the model which was trained on the NAB dataset and tested on live online retail systems to enhance decision-making, incident handling and business continuity. However, due to excessive computational requirements, time delays that may occur during blockchain consensus, and a reliance of high-quality training data to sustain performance in different environments, means that these limitations need to be highlighted; all of which forms part of a deeper understanding of the risk meeting the reality.

In [10] presented a methodology for detecting anomalies in cloud security that combines deep learning and blockchain. The technique examines systems logs, finds anomalies, and predicts cloud attacks using Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN), enhances blockchain-based cloud deployments, minimizes feature similarity and employs QoS-aware sidechains. Blockchain prevents manipulation by ensuring log integrity and transparency. Precision, accuracy, recall, AUC, block mining latency and throughput are the metrics of the system. It has real time detection, better security and accuracy, but it has challenges of processing complexity, energy consumption, and latency (due to blockchain) when used cloud environments that are dynamic.

An innovative Multi-layered Blockchain Security Framework (M-BSF) was presented [11] in 2023 to safeguard Privacy-Enabled (PE) Internet of Things (IoT) data at the edge, fog, and cloud layers. At the edge and fog levels, the approach utilized public key encryption (Baby Kyber) and private key decryption (Scaling Kyber) algorithms. The data on the blockchain stores secured information at the cloud layer using the Argon2di hashing algorithm [14 – 17].

A. Problem statement

The continuous proliferation of cloud computing and Internet of Things (IoT) architectures has concomitantly strained the triad of data protection availability, confidentiality, and integrity beyond the capacities of legacy practices. Although established cryptographic procedures retain applicability within certain cloud modalities, their robust application is undermined by emergent quantitive threats, namely quantum adversaries capable of efficiently dismantling prevailing algorithms. Centralized repositories, ubiquitous in contemporary cloud ecosystems, introduce systemic vulnerabilities characterised by monopoints of failure, the potential for unmediated authorizations, and the concomitant leakage of sensitive information. Furthermore, prevailing security architectures, designed for more static and compartmental environments, lack the agility needed for the massive, fluid, and heterogeneous workloads characteristic of modern cloud services. Addressing these challenges therefore necessitates the engineering of cryptographic paradigms simultaneously advanced, scalable, and, wherever feasible, resistant—not merely tolerant—of quantum incursions, while embracing a decentralised architecture to guarantee persistent, non-reputable data security.

III. PROPOSED METHODOLOGY

A new Multi-Channel Quantum Convolutional Neural Network with Walk-Spread Algorithm (M-CQCNNet-WSA) is introduced for secure and scalable cloud computing IoT data management. The workflow process starts with collecting data from different IoT devices and follows with an encryption store using the McEliece Post-Quantum Cryptosystem, rendering data useful and quantum-resistant. The encrypted data is analyzed using the M-CQCNNet, which identifies the anomalies detected from multiple channels of data and optimizes neural network weights for detection accuracy and computing performance with the Walk-Spread Algorithm (WSA). Eventually, the processed data is secured on a Lightweight Blockchain by using the Practical Byzantine Fault Tolerance Consensus (PBFT-CLBC), providing a decentralized, tamper-proof and transparent storage. By leveraging quantum-secure encryption, intelligent anomaly detecting, and blockchain-based storage, M-CQCNNet-WSA provides an end-to-end cryptographic security architecture for IoT based cloud systems, with appropriate and relevant provision of addressing new potential hazards and threats, with scalable data integrity and privacy. The proposed blockchain based M-CQCNNet-WSA architecture is illustrated in figure 1.

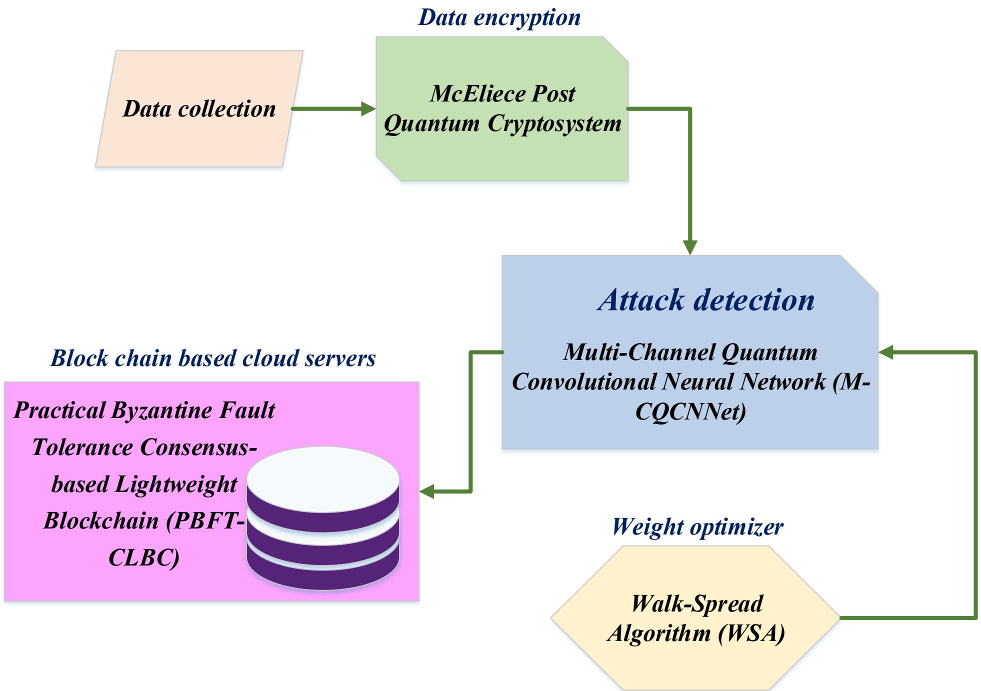

Fig.1. Proposed blockchain based M-CQCNNet-WSA architecture

A. Data Acquisition

Initially, IoT devices, such as sensors, actuators, detectors, and regulators, ingest data from multiple places at once. These devices use encryption transmission methods to promote secure connectivity to the IoT servers in the Cloud, thereby improving the iot data integrity more rapidly. The McEliece Post Quantum Cryptosystem was used as an additional security method prior to changing the plain message to this format for transmission.

B. Encryption using McEliece Post Quantum Cryptosystem

Protected, the acquired information is used through a McEliece Post-Quantum Cryptosystem (PQC) [12]. This cryptographic protocol is still safe from quantum computers. Unlike the traditional schemes based on the problems of factorisation or discrete logarithm, the McEliece system is based on the error-correcting codes. Subsequently, adversaries including those with quantum technology cannot break it computationally. Thus, McEliece is a powerful tool to enhance the security of storing information in the cloud:

Step 1: Key Generation Request

The encryption process produces a set of public keys, that are composed of functions Scrambler Matrix (t), Generator Matrix (h), and Permutation Matrix (q). In order to find this out, the system deliberately adds false bits in the ciphertext before transmission. These errors can only be verified by the concerned party by using the private key which only the recipient has. This design guarantees that the communication will be confidential as well as complete.

Step 2: Verification and Secure Setup

Each constituent node within the cloud architecture undergoes stringent authentication predicated upon a zero-trust security paradigm, thereby obviating any implicit trust regardless of the node's geographical or topological placement within the network. Authentication is facilitated exclusively by services possessing the minimal set of privileges required to execute the key-generation protocol, ensuring that the security and cryptographic integrity of cloud-resident data are preserved at all times.

Step 3: Key Splitting and Distribution

Upon verification, the private key components are separated into smaller submatrices: the Generator Matrix (h), the Permutation Matrix (q) and the Scrambler Matrix (t). The orchestrating service retains some of those, while the others are distributed across multiple cloud nodes to allow for parallel processing. When using cloud systems, the

979-8-3315-9929-4/25/$31.00 ©2025 IEEE

ability to split these results in the potential optimization of resources and faster processing.

Step 4: Parallel Computation Using Cloud Resources

Every cloud node works in a divide-and-conquer manner to independently compute the individual submatrix it has allocated. Large matrices can be divided into many smaller tasks to allow nodes to perform calculations properly. When the calculations complete, the orchestrator service receives a partial result. By using this distributed method, energy usage is maximized through use of cloud scalability and latency is reduced.

Step 5: Key Reconstruction and Public Key Formation

The orchestrator service uses matrix multiplication to recreate the entire public key after aggregating the calculated sub matrices equation (1).

$$h_{pub} = t \times h \times q \quad (1).$$

The final public key ensures quick, scalable, and extremely secure encryption by acting as the cornerstone of quantum-resistant secure communication within the cloud infrastructure. The key generation process is described in algorithm 1.

Algorithm 1: key generation

block size d ($^\wedge 2$), number of levels m

Random generator matrix h

Function $divide\ and\ conquer(nu)$:

  If $\leftarrow 0\$$ ***then***

    Return random $d \times d$ matrix nu

  end

return

else

  $nu_1 \leftarrow generatematrix[k-1]$

  $nu_2 \leftarrow generatematrix[k-1]$

  Divide nu_1 and nu_2 into $d \times d$ sub-matrices

end

For $l \in range(d^k)$ ***do***

  For $o \in range(d^k)$ ***do***

    $M_{l,o} \leftarrow random d \times d\,(nu)$

  end

end

$q_{l,o} \leftarrow M_{l,o} nu_1 + M_{l,m+d^j} nu_2$ for $1 \le l, o \le d^k$

Combine $q_{l,o}$ into a single $d^k \times d^k$ matrix nu

Return nu

Step 6: Secure Communication using Quantum-Resistant Keys

Cloud services encrypt data communications between clients and backend services once the post-quantum public key has been successfully produced. This encryption protects critical cloud data from potential risks from quantum computing by ensuring confidentiality, integrity, and authenticity. Continuous monitoring of encrypted data ensures proactive protection and upholds the cloud environment's overall security posture by spotting any possible threats or irregularities. The intended recipient securely decrypts the encrypted data using the matching private key components after receiving it, allowing only authorized people to access the original contents.

C. Threat Detection using Multi-Channel Quantum Convolutional Neural Network (M-CQCNNet)

Multi-Channel Quantum Convolutional Neural Network (M-CQCNNet) [13] uses a three-stage architecture that includes: (i) multi-channel data uploading, (ii) quantum convolution, and (iii) data reconstruction to process the encrypted data for attack detection. With the help of quantum circuits, M-CQCNNet can analyze threats quickly and securely by operating directly on encrypted inputs.

A. Multi-Channel Data Uploading

A quantum circuit receives encrypted data with a three-dimensional structure $J \in \Re^{D_{in} \times I \times X}$ via a multi-channel encoding procedure. Each value from the input tensor is converted into a quantum state by a group of rotation gates, such as S_y, S_z and S_a . These gates enable the representation of channel and spatial dimensions in quantum superposition by acting on individual qubits. The encoding for a specific geographic position j is expressed by equation (2).

$$|\psi\rangle_j = \prod_{d=0}^{D_{in}} \prod_{k=0}^{i_d x_d} S_\vartheta(y_{d,k}) |0\rangle^{\otimes r} \quad (2).$$

In this case, r is the number of qubits allotted per quantum for encoding, $y_{d,k}$ indicates the quantum value in channel d at position k , S is a rotation operator parameterized by angle depending on the encoding scheme in ϑ learnable weight function, and $i_d x_d$ indicates the tensor product representing the combined quantum state of multiple qubits.

b. Quantum Convolution

Once the quantum states are encoded, they are convolutionized using a parameterized Variational Quantum Circuit (VQC). The VQC is composed of trainable quantum gates, like rotation and entangling gates, which change the quantum state to extract useful characteristics. These processes are used in place of conventional convolution techniques to enable high-dimensional pattern recognition in the encrypted domain. For each point j , the output of this stage is obtained by quantum measurement in the Pauli-Z basis, as explained by equation (3).

$$\langle P\rangle \stackrel{\Delta}{=} \left\{ \langle P|(\psi_\theta)_j \rangle | j \in N[0, I_{out} \times X_{out}] \right\} \quad (3).$$

where, P is the measurement operator (often a Pauli-Z operator), and $(\psi_\theta)_j$ is the quantum state at location j following the VQC. With r being the number of qubits measured per place and the expectation value ranging from -1 to 1, a vector of expectation values representing extracted features is produced.

c. Data Reconstruction

For further processing and reconstruction, the measured expectation values are vectors of classical data that are placed into a thick, fully linked layer. The dimensions of this layer are (r, D_{out}), where r is the number of measured qubits per spatial position and D_{out} is the number of output channels or feature maps the model produces after quantum convolution. The thick layer transforms the vector of quantum observations into a new feature space, as explained by equation (4).

$$g(\langle P\rangle) \in \Re^{(I_{out} \times X_{out}) \times D_{out}} \quad (4).$$

where, g indicates the fully connected transformation, and I_{out} and X_{out} are the normal and attack of the output feature maps, respectively. The output is reshaped to create the reconstructed tensor. The variational quantum signatures extracted and stored in the enigmatic input-hologram manifest through embedded feature maps, which, following the quantum-to-classical transduction, are ingested by dense feed-forward architectures designed to extricate and magnify signs of hostile intent.

D. Weight optimizer using Walk-Spread Algorithm (WSA)

The Weight Optimisation Unit of the M-CQCNNet employs the Walk-Spread Algorithm (WSA) to synthesise the tensions between broad discovery and radical refinement within a self-regulating, translucent apparatus. Initialisation is realised through a spatially-extended population of weight candidate vectors, uniformly dispersed across the weight manifold; iterative elevation transpires beneath two harmonised fronts: a stochastic expansion kernel that interrogates discrete Hessian neighbourhoods and a structured trans-location that drags candidate vectors toward the identified minima. During trans-location, candidate neighbourhoods, assessed by ordinal attractiveness, provide poised pivots that trap proximal populations, mediated by inner- and own-attractors that pull discipline and order. In conjugate expansion, a surging radius of neighbourhoods random-walks reactively samples mesoscopic regions and materialises a spatially unravelling manifold of sampled perturbations. When faced by fractal rugged landscapes, the M-CQCNNet prosecute global meta-interiors concentrated in pivotal neighbourhood weight estates; meta-trials are reiterated by a dominical moderation that enforces contracted fitness ascent founded on structured modelling of visit history, thus guaranteeing that the fractal escape channels transverse lower local basins and that perturbational cycles asymptote toward cumulatively superior refinements.

Step 1: Initialization

Initially, a random population of unit's potential solutions is generated across the designated search space. The multidimensional solution space, where each unit $v_{j,k}$ is represented as a vector, is described by equation (5).

$$v_{j,k} = cm_k + s(0,1) \times cx_k \quad (5).$$

Step 2: Fitness calculation

The effectiveness of a given weight vector is quantified by a specialized fitness-responsive assessment. The task is to ascertain the weight arrangement that yields the least value of the predetermined target metric, in accordance with the paradigm articulated in equation (6).

$$Fitness = f \min(\vartheta) \quad (6).$$

where, the fitness value is denoted by f. The weight vector is denoted by ϑ.

IV. RESULTS

A. Dataset description

To validate our proposed model, this study evaluate its privacy and protection policies using the NAB dataset, a real-life time-series dataset for anomaly detection. This dataset includes a variety of sources, including sensors, servers, and social networks and demonstrates high variance between each source, which is typical of an IoT environment. The dataset includes both normal and anomalous data, making it robust to evaluate the model's ability to identify anomalies. There are a total of 345,444 samples in the dataset; use 80% for training and 20% for testing.

B. Performance analysis

The Blockchain-based M-CQCNNet-WSA paradigm integrates its cutting-edge anomaly detection with quantum-resistant encryption for secure and scalable IoT data handling in cloud computing. The performance of the proposed M-CQCNNet-WSA on a blockchain is evaluated with respect to accuracy, precision, F1-score, recovery efficiency, encryption time, and decryption time. Current techniques such as MBRA [8], GAN [9], CNN [10], and M-BSF [11] are used to benchmark the model.

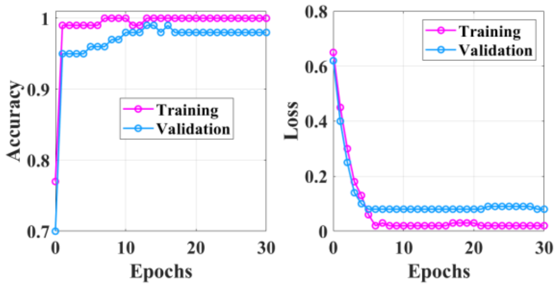

Fig.2. Training and Validation Performance of M-CQCNNet-WSA

The proposed M-CQCNNet-WSA model's accuracy and loss trends across 30 epochs are shown in Figure 2. The validation accuracy (blue) stabilizes at about 98.5%, while the training accuracy (magenta) quickly exceeds 99.9%, demonstrating outstanding generalization. The validation loss drops from 0.64 to almost 0.06 on the right, and the training loss drops from 0.65 to almost 0.01. The model's excellent anomaly detection accuracy and low error rate are demonstrated by these results, which also demonstrate how well the Walk-Spread Algorithm (WSA) optimizes learning for safe IoT-cloud contexts.

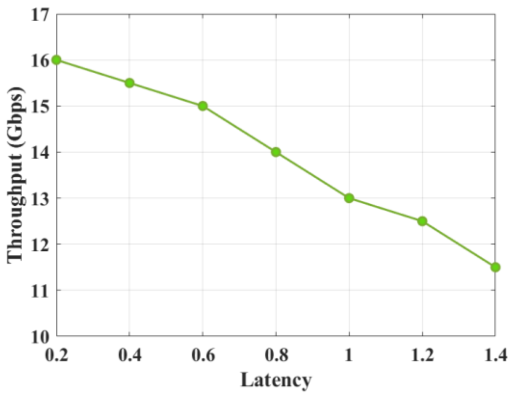

Fig.3. Latency vs. Throughput Performance of M-CQCNNet-WSA

The M-CQCNNet-WSA framework's inverse latency-throughput relationship is depicted in Figure 3. During the period between 0.2 and 1.4 seconds, throughput drops from 16 Gbps to 11 Gbps. The throughput specifically decreases from 16 Gbps at 0.2s latency to 15.5 Gbps at 0.4s, 15 Gbps at 0.6s, 14 Gbps at 0.8s, 13 Gbps at 1.0s, 12.5 Gbps at 1.2s, and 11 Gbps at 1.4s. According to these findings, M-CQCNNet-WSA maintains a high throughput in low-latency scenarios, enabling safe real-time IoT data handling in cloud systems.

TABLE 1. PERFORMANCE COMPARISON

Method	Accuracy (%)	Precision (%)	F1-Score (%)	Restoring Efficiency (%)	Encryption Time (ms)	Decryption Time (ms)
MBRA [8]	91.35	89.72	90.08	88.93	4.85	4.91
GAN [9]	93.11	91.25	92.02	90.74	4.12	4.20
CNN [10]	94.60	93.18	93.64	92.51	3.76	3.81
M-BSF [11]	95.77	94.92	95.11	94.68	3.33	3.40
Proposed	99.28	99.05	99.11	98.94	2.05	2.08

The success of various approaches across important evaluation indicators is highlighted in the performance comparison table 1. MBRA [8] has encryption and decryption speeds of 4.85 ms and 4.91 ms, respectively, and achieves 91.35% accuracy, 89.72% precision, 90.08% F1-score, and 88.93% restoration efficiency. With 93.11% accuracy, 91.25% precision, 92.02% F1-score, and 90.74% restoring efficiency, GAN [9] outperforms this. Results are further improved by CNN [10] to 94.60% accuracy and 93.64% F1-score. M-BSF [11] achieves 95.11% F1-score and 95.77% accuracy. With the lowest encryption (2.05 ms) and decryption (2.08 ms) timings, 99.28% accuracy, 99.05% precision, 99.11% F1-score, and 98.94% restoration efficiency, the suggested Blockchain-based M-CQCNNet-WSA performs noticeably better than the others.

V. CONCLUSION

The architecture hereby delineated—Multi-Channel Quantum Convolutional Neural Network fortified by Walk-Spread Algorithm (M-CQCNNet-WSA)—demonstrates an extraordinary capability for the secure supervision of Internet of Things (IoT) telemetry transmitted across cloud environments, whilst preserving the robustness of anomalous-event identification. Empirical assessment yields an overall accuracy of 99.28%, precision of 99.05%, and an F1-score of 99.11%, confirming the design's efficacy. The overall topology leverages the McEliece post-quantum cryptosystem, whose theoretically enduring polynomial-code encoding confounds anticipated quantum cryptanalytic efforts, further strengthened by the Walk-Spread Algorithm, which constrains inferential bandwidth through an encased compressive-detection prism. Beneath this architecture, a Practical Byzantine Fault Tolerance ledger coalesced with Composite Lifetime-based Charge protocols (PBFT-CLBC) yields a decentralised, tamper-evident vault marked by persistently unveiled provenance. The resultant system confers quantum-resilient encryption parameterised by time, heightened fidelity of intrusion detection, microsecond region latencies for both encryption and decryption, and linear economics of scalability across the distributed vault. Nonetheless, the prototype datasets currently under scrutiny—non-corporate by design—incorporate exceptionally extensive, heterogeneously-structured, and semi-volatile records, and the concomitant emergent enhancements of auto-scaled, dynamic resource allocation have restricted the system's agility in adapting to the real-time procedural fluctuations intrinsic to contemporary IoT constellations. As temporarily facilitating schemes for enhancement, the forthcoming model synthesis will recalibrate operational genus via successive, auto-scaling epochs that juxtapose prescribed metrics with acceptance parameters dictated by the user—namely, pre-specified, regionally modifiable thresholds, and analogous heuristics. Simultaneously, intelligent performance metrics will be subjected to progressive, extending-user-character incorrect, federated datasets. The amendments shall substantively augment functional heterogeneity, elevate operational throughput, and reinforce systemic resilience across the temporally discontinuous IoT-cloud choreography, wherein both predictive and reactive fluctuations are indeterminate yet demonstrably ordinal.

REFERENCES

[1] N. Koroniotis, N. Moustafa, and J. Slay, "A new intelligent satellite deep learning network forensic framework for smart satellite

networks," Comput. Electr. Eng., vol. 99, p. 107745, 2022. DOI: 10.1016/j.compeleceng.2022.107745

[2] Kolli, Nagaraju. "Navigating the Cloud: Security, Compliance, and Risk Challenges in SME Adoption." Journal of Information Technology and Digital World 7, no. 3 (2025): 200-215 https://doi.org/10.36548/jitdw.2025.3.001

[3] Z. Ullah, B. Raza, H. Shah, S. Khan, and A. Waheed, "Towards blockchain-based secure storage and trusted data sharing scheme for IoT environment," IEEE Access, vol. 10, pp. 36978–36994, 2022. DOI: 10.1109/ACCESS.2022.3164081

[4] A. Padma and M. Ramaiah, "Blockchain based an efficient and secure privacy preserved framework for smart cities," IEEE Access, vol. 12, pp. 21985–22002, 2024. DOI: 10.1109/ACCESS.2024.3364078

[5] W. Zhao, L. Wang, Z. Zhang, H. Fan, J. Zhang, S. Mirjalili, et al., "Electric eel foraging optimization: A new bio-inspired optimizer for engineering applications," Expert Syst. Appl., vol. 238, p. 122200, 2024. DOI: 10.1016/j.eswa.2023.122200

[6] X. Jin, J. Zhang, J. Kong, T. Su, and Y. Bai, "A reversible automatic selection normalization (RASN) deep network for predicting in the smart agriculture system," Agronomy (Basel), vol. 12, no. 3, p. 591, 2022. DOI: 10.3390/agronomy12030591

[7] Ramkumar, M. S., Subramani, J., Sivaramkrishnan, M., Munimathan, A., Michael, G. K. O., & Alam, M. M. (2025). Optimal energy management for multi-energy microgrids using hybrid solutions to address renewable energy source uncertainty. *Scientific Reports*, *15*(1), 7755.

[8] Z. Ullah, B. Raza, H. Shah, S. Khan, and A. Waheed, "Towards blockchain-based secure storage and trusted data sharing scheme for IoT environment," IEEE Access, vol. 10, pp. 36978–36994, 2022. DOI: 10.1109/ACCESS.2022.3164081

[9] R. R. Irshad, S. Hussain, I. Hussain, J. A. Nasir, A. Zeb, K. M. Alalayah, et al., "IoT-enabled secure and scalable cloud architecture for multi-user systems: A hybrid post-quantum cryptographic and blockchain-based approach toward a trustworthy cloud computing," IEEE Access, vol. 11,105479–105498, 2023. DOI: 10.1109/ACCESS.2023.3318755

[10] K. Zkik, A. Belhadi, S. Kamble, M. Venkatesh, M. Oudani, and A. Sebbar, "Cyber resilience framework for online retail using explainable deep learning approaches and blockchain-based consensus protocol," Decis. Support Syst., vol. 182, p. 114253, 2024. DOI: 10.1016/j.dss.2024.114253

[11] Sivaramkrishnan, M., Ancelin, L., Ramkumar, M. S., Al Jawad, O. J. J. A., Giri, J., & Vignesh, T. (2025, May). Neural Architecture Search-Guided Physics-Informed Neural Network for Energy Management in Hybrid Energy Storage System with Electric Vehicles. In *2025 7th International Conference on Inventive Material Science and Applications (ICIMA)* (pp. 552-557). IEEE.

[12] A. V. Nagarjun and S. Rajkumar, "Design of An Anomaly Detection Framework For Delay & Privacy-Aware Blockchain-Based Cloud Deployments," IEEE Access, vol. 12, pp. 84843–84862, 2024. DOI: 10.1109/ACCESS.2024.3414998

[13] C. Awasthi, P. K. Mishra, P. K. Pal, S. B. Khan, A. K. Agarwal, T. R. Gadekallu, et al., "Preservation of sensitive data using multi-level Blockchain-based secured framework for edge network devices," J. Grid Comput., vol. 21, no. 4, p. 69, 2023. DOI: 10.1007/s10723-023-09699-2

[14] M. Ngouen, M. A. Rahman, N. Prabakar, S. Uluagac, and L. Njilla, 2023, October. Q-SECURE: a quantum resistant security for resource constrained IoT device encryption. In 2023 10th International Conference on Internet of Things: Systems, Management and Security (IOTSMS) (pp. 141-148). IEEE. DOI: 10.1109/IOTSMS59855.2023.10325770

[15] E. J. Roh, J. Y. Shim, J. Kim, and S. Park, "Hybrid quantum-classical 3D object detection using multi-channel quantum convolutional neural network," J. Supercomput., vol. 81, no. 3, pp. 1–24, 2025. DOI: 10.1007/s11227-025-06968-7

[16] P. D. Kusuma and A. L. Prasasti, "Walk-Spread Algorithm: A Fast and Superior Stochastic Optimization," International Journal of Intelligent Engineering & Systems, vol. 16, no. 5, 2023.

[17] J. Qi and Y. Guan, "Practical Byzantine fault tolerance consensus based on comprehensive reputation," Peer-to-Peer Netw. Appl., vol. 16, no. 1, pp. 420–430, 2023. DOI: 10.1007/s12083-022-01408-2